

13 September 2026

Extension of the v5.6 peak catalogue

Conditional upper-limit echoes from projected 2021 peaks

This extension follows the v5.6 injected peaks across the mass scan to predict nearby upper-limit dips, or *echoes*. It reuses all 300 saved toys across 15 scenarios, with separate nominal $1\% \times 100$ and $10\% \times 10$ projections.

The local-significance scaling that motivates the injections

For unchanged selection and shapes, a persistent excess dominated by counting statistics scales with the exposure multiplier $k = 1/f$ as

$$S_{100\%} = kS_f, \quad B_{100\%} = kB_f, \quad Z_{100\%} \simeq \frac{kS_f}{\sqrt{kB_f}} = \sqrt{k} Z_f, \quad (1)$$

$$\boxed{Z_{1\% \rightarrow 100\%}^{\text{target}} = 10Z_{1\%}} \quad \boxed{Z_{10\% \rightarrow 100\%}^{\text{target}} = \sqrt{10} Z_{10\%}}.$$

The injected amplitude was matched to this fixed-mass Asimov target. This is a conditional construction, not a bound on future significance; a statistical fluctuation need not persist as a signal.

A conditional upper-limit curve for each saved spectrum

At each mass m , a bin-integrated Gaussian count template $t_i(m)$ is fitted to Poisson counts with a profiled GP-background constraint:

$$\ell(A, \theta; m) = \sum_i [n_i \ln \lambda_i - \lambda_i] - \frac{1}{2} \theta^T \theta, \quad \lambda_i = b_i + (L\theta)_i + A t_i(m). \quad (2)$$

The sideband-conditioned b and LL^T are recomputed at every tested mass. Writing $\hat{A}_+ = \max(0, \hat{A})$, with the nuisance profiled there,

$$\tilde{q}_A = \begin{cases} -2 \ln \frac{L(A, \hat{\theta}_A)}{L(\hat{A}_+, \hat{\theta}_+)}, & \hat{A} \leq A, \\ 0, & \hat{A} > A, \end{cases} \quad \text{CL}_s(A) = \frac{\Pr_A(\tilde{q}_A \geq \tilde{q}_A^{\text{obs}})}{\Pr_0(\tilde{q}_A \geq \tilde{q}_A^{\text{obs}})}. \quad (3)$$

The bounded asymptotic tails use the profiled background Asimov response; A_{90} solves $\text{CL}_s(A_{90}) = 0.10$ [1]. These are conditional, simulated, observed-like 90% CL_s curves.

Predicting the positions and depths of the echoes

Let $A_{90}^{(b)}(m)$ be the result from the same scenario's generator continuum alone. The reference and injected spectra are each reanalysed on the full grid. Define a count-limit ratio and the depth of its selected flank minimum by

$$R(m) = \frac{A_{90}^{\text{injected}}(m)}{A_{90}^{(b)}(m)}, \quad D = 1 - R(m_{\text{echo}}), \quad m_{\text{echo}} \in m_0 \pm [2, 8] \sigma_m(m_0). \quad (4)$$

Each flank is clipped to the saved scan support. The deepest local minimum on the saved scan lying in that flank is selected; if none exists, the flank minimum is retained and marked. $D > 0$ means a tighter limit than the background reference. The contiguous saved-grid interval around the selected minimum with $R \leq 0.90$ records a depression of at least 10%, when present. All choices are declared before examining observed full-statistics data.

Why a strong peak can leave a negative fitted structure nearby

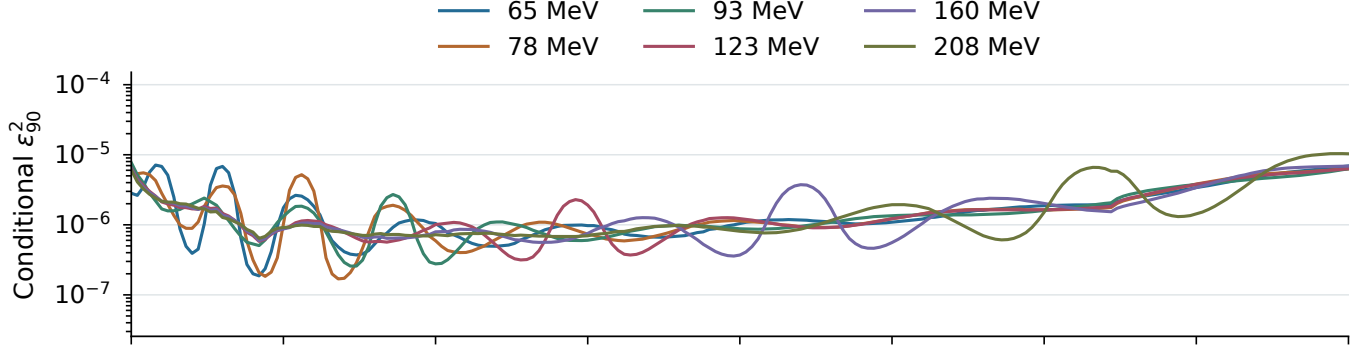
As the excluded fit window moves away from an injected peak, signal bins and tails can enter the GP training sidebands. Their influence on the predicted continuum can produce negative fitted amplitudes at nearby masses and tighten the upper limit. The signed profile score $r = \text{sign}(\hat{A}) \sqrt{2(\ell_{\text{free}} - \ell_{A=0})}$ makes that response visible. The one-sided discovery score would set negative values to zero. Echoes are conditional responses of this model to the injected structure; real detector and background effects could produce different shapes. Generated counts and fitted total expectations remain positive.

No observed 100% 2021 histogram is used. Neither these simulated limits nor the 20-toy spreads establish calibrated coverage, a probability of real echoes, or an unblinding decision. Nearby and overlapping scenarios are alternative injections and must not be combined.

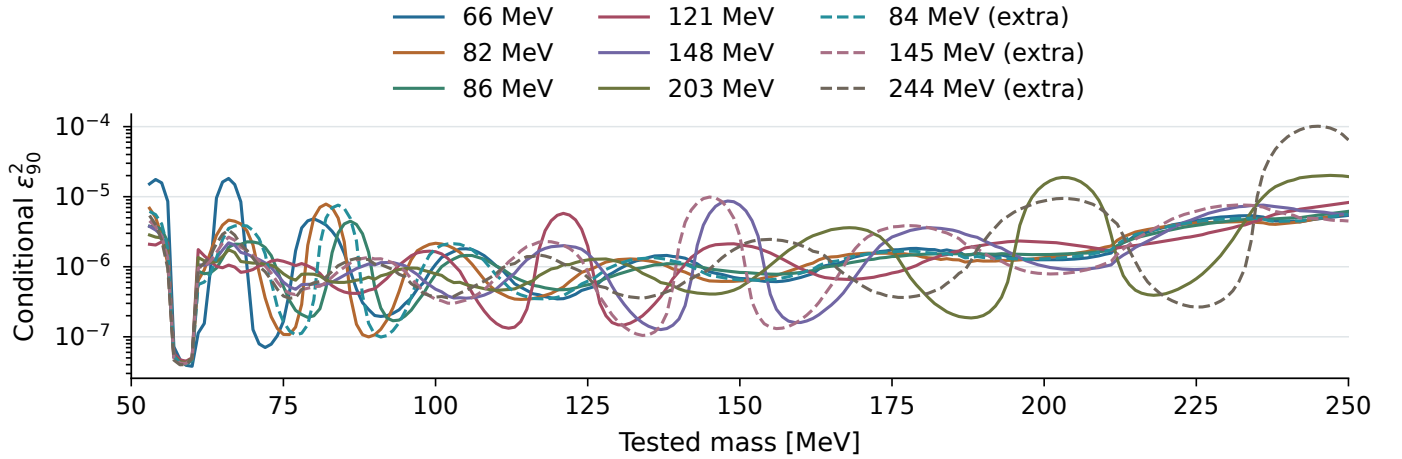
Full-range upper-limit projections

The curves below show each target-matched Asimov spectrum. The upper panel compares the six requested regions using the native 10% source; the lower panel shows the historical nominal 1% alternatives. Labels give the actual injected masses. Each scenario retains its own source-conditioned generator continuum, so the reference curve also depends on the injection location.

10% x10: matched-Asimov injections at the labelled masses



1% x100: matched-Asimov injections at the labelled masses



Count limits and the prompt-density conversion

The displayed coupling coordinate uses the standard conversion separately for each tested spectrum n :

$$K_n(m) = \frac{3\pi m f_{\text{rad}}^{\text{eff}}}{2\alpha} \rho_n(m), \quad \epsilon_{90,n}^2(m) = \frac{A_{90,n}(m)}{K_n(m)} B_e^{-1}(m). \quad (5)$$

Here $f_{\text{rad}}^{\text{eff}} = 0.0477$, $\alpha = 1/137$, and ρ_n is the count density from fractional bin overlap within $\pm 1.64\sigma_m$; masses and density use GeV consistently. Above $2m_\mu = 211.316749$ MeV, the inherited visible branching correction is $B_e^{-1} = 1 + \sqrt{1 - 4u}(1 + 2u)$, $u = (m_\mu/m)^2$; below threshold it is one. This explains a physical change in the conversion near 211 MeV.

The echo ledger instead uses $A_{90}/A_{90}^{(b)}$. This is exactly the ratio of coupling limits evaluated with the same frozen background density in both numerator and denominator, isolating the fit response from the small change in the density normalization. Both conversion variants are saved.

Predicted echo locations and depths

The table records the two selected minima of each matched-Asimov limit ratio. Depth is relative to the scenario's background-only Asimov count limit at the same mass. The listed interval contains saved 1 MeV hypotheses with $R \leq 0.90$ connected to that minimum.

Option	m_0 [MeV]	Left m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Right m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]
1% $\times$ 100	66	61 [†]	92.9	57–61	72	92.2	71–76
10% $\times$ 10	65	60 [†]	79.2	58–60	70 [†]	74.0	70–73
1% $\times$ 100	82	75	84.9	71–77	89	86.7	87–94
10% $\times$ 10	78	72	77.7	69–73	84	78.2	83–88
1% $\times$ 100	86	79	76.7	75–81	93	76.0	91–99
10% $\times$ 10	93	86	62.5	83–88	100	65.2	98–105
1% $\times$ 100	121	112	81.1	105–115	131	80.5	127–139
10% $\times$ 10	123	115	53.3	109–117	132	54.3	129–138
1% $\times$ 100	148	137	83.7	129–140	160	84.2	156–170
10% $\times$ 10	160	149	60.4	142–152	172	61.1	168–180
1% $\times$ 100	203	189	86.8	177–192	219	86.6	214–232
10% $\times$ 10	208	194	60.8	185–197	224	61.2	219–234
1% $\times$ 100	84	78	85.9	73–79	91	86.1	89–96
1% $\times$ 100	145	135	85.7	126–138	156	86.9	152–166
1% $\times$ 100	244	227	92.2	214–230	–	–	–

†: selected point is a search-flank or scan endpoint. *: no local minimum; the fallback flank minimum is reported. A dash means an unavailable flank or no connected $R \leq 0.90$ interval. The 244 MeV injection's right search flank lies outside the scan. A positive depth identifies a depression; a nonpositive depth does not. The additional 1% injections at 84, 145 and 244 MeV follow the six requested pairs.

The saved spectra and source interpretation

Every scenario contains a background-only continuum, a target-matched injection, a simple yield-scaled injection and the same 20 toys used in v5.6.0. Only the target-matched injection has toys. The simple yield-scaled amplitude is $a_{\text{yield}} = k \max(\hat{a}_f, 0)$; the matched amplitude was chosen so that its fixed-mass Asimov score equals $\sqrt{k}Z_f$. Agreement with that target is an injection construction, not evidence that the GP satisfies ideal exposure scaling. The historical 1% input is the last-used study source, with its own reviewed kernel coordinates and 40–300 MeV conditioning support. The native 10% option uses its own reviewed states and 36–300 MeV support. The respective limit grids are 53–250 and 50–250 MeV in 1 MeV steps. The 1% lower-edge control region at 50–52 MeV remains excluded. Both use the inherited fully scaled resolution, 0.625 MeV bins and a $\pm 2.25\sigma_m$ fit exclusion. The exact selection, trigger-category membership, overlap and effective exposure equivalence of the two source samples remain unverified. The source significances inherited from v5.6 are fresh evaluations at archived kernels, rather than verbatim older released values. Numerical-method differences must not be read as exposure effects. The 1% injections at 86 and 148 MeV are inherited source-region boundary controls, not interior observed peaks. In particular, the 1% 86 MeV control is distinct from the native 10% injection at 93 MeV.

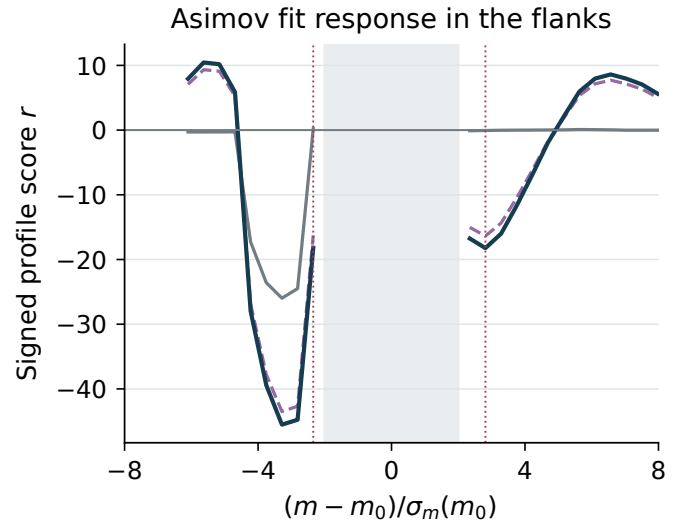
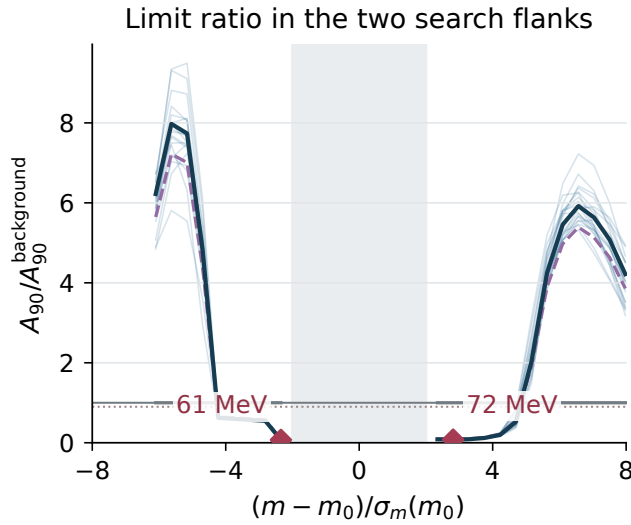
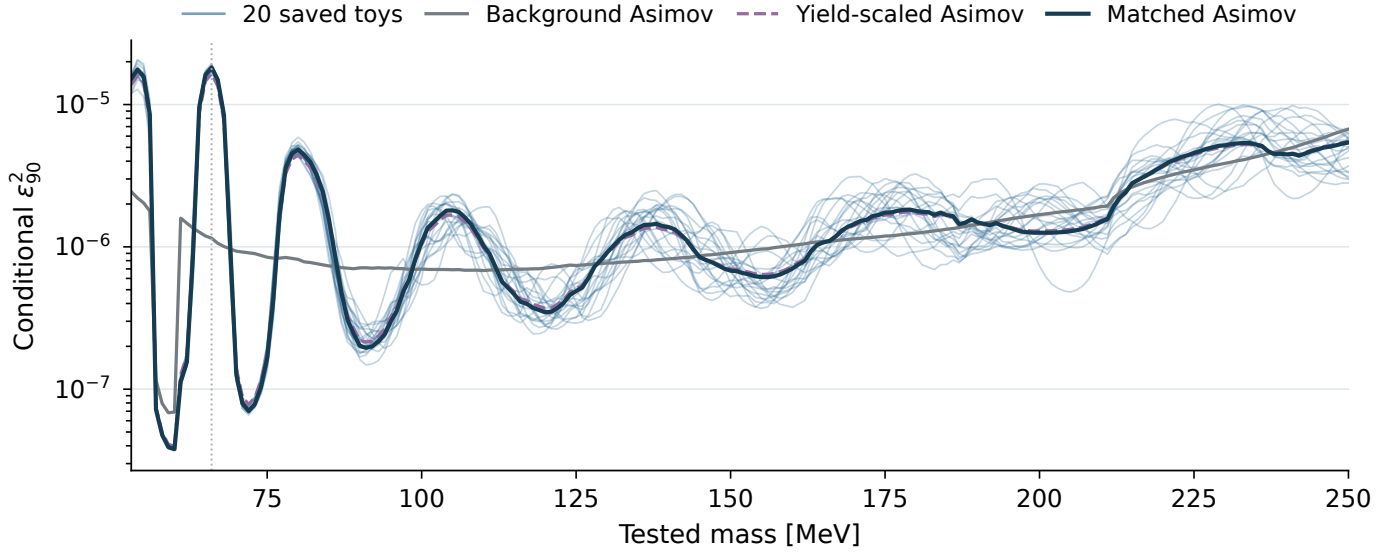
What the toy summaries mean

On each flank, the toy minimum is searched over the same declared interval. The following pages report empirical medians and 16–84% ranges for its mass. To test persistence at a common location, the ratio summary and threshold count use each toy's ratio at the fixed matched-Asimov echo mass, not at its moving minimum. The plots contain every saved toy curve. These are descriptive summaries of 20 conditional realizations. They are not a background expected band, a simultaneous uncertainty band, or a calibrated confidence interval for the echo.

65 MeV region: $1\% \times 100$

$m_0 = 66$ MeV, $\sigma_m = 2.132$ MeV; $Z_{\text{target}} = 26.12$, $a_{\text{match}}/a_{\text{yield}} = 1.115$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	61 [†]	92.9	57–61	61.0 [61.0, 61.0]	0.07 [0.07, 0.08]	20/20
Right	72	92.2	71–76	72.0 [72.0, 72.0]	0.08 [0.08, 0.08]	20/20

Search bounds after clipping to the scan are left: 53.00–61.74 MeV; right: 70.26–83.05 MeV. [†] marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

$m_0 = 65$ MeV, $\sigma_m = 2.122$ MeV; $Z_{\text{target}} = 7.58$, $a_{\text{match}}/a_{\text{yield}} = 1.120$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Figure 10 consists of three panels. The top panel is a line plot showing the conditional variance of the Asimov fit, ϵ_{90}^2 , on a logarithmic y-axis (ranging from 10^{-6} to 10^{-5}) versus the tested mass m [MeV] on the x-axis (ranging from 50 to 250). The plot shows multiple orange lines representing 20 saved toys, a grey line for the Background Asimov, a purple line for the Yield-scaled Asimov, and a dark blue line for the Matched Asimov. The Matched Asimov line shows a clear dip around 70 MeV. The bottom left panel is a line plot showing the limit ratio in the two search flanks, $A_{90}/A_{90}^{\text{background}}$, on the y-axis (ranging from 0 to 4) versus $(m - m_0)/\sigma_m(m_0)$ on the x-axis (ranging from -8 to 8). The plot shows multiple orange lines for 20 saved toys, a grey line for Background Asimov, a purple line for Yield-scaled Asimov, and a dark blue line for Matched Asimov. The Matched Asimov line shows a clear dip around 70 MeV. The bottom right panel is a line plot showing the signed profile score r on the y-axis (ranging from -6 to 4) versus $(m - m_0)/\sigma_m(m_0)$ on the x-axis (ranging from -8 to 8). The plot shows multiple orange lines for 20 saved toys, a grey line for Background Asimov, a purple line for Yield-scaled Asimov, and a dark blue line for Matched Asimov. The Matched Asimov line shows a clear dip around 70 MeV.

Top: full-range coupling limits with each spectrum’s own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

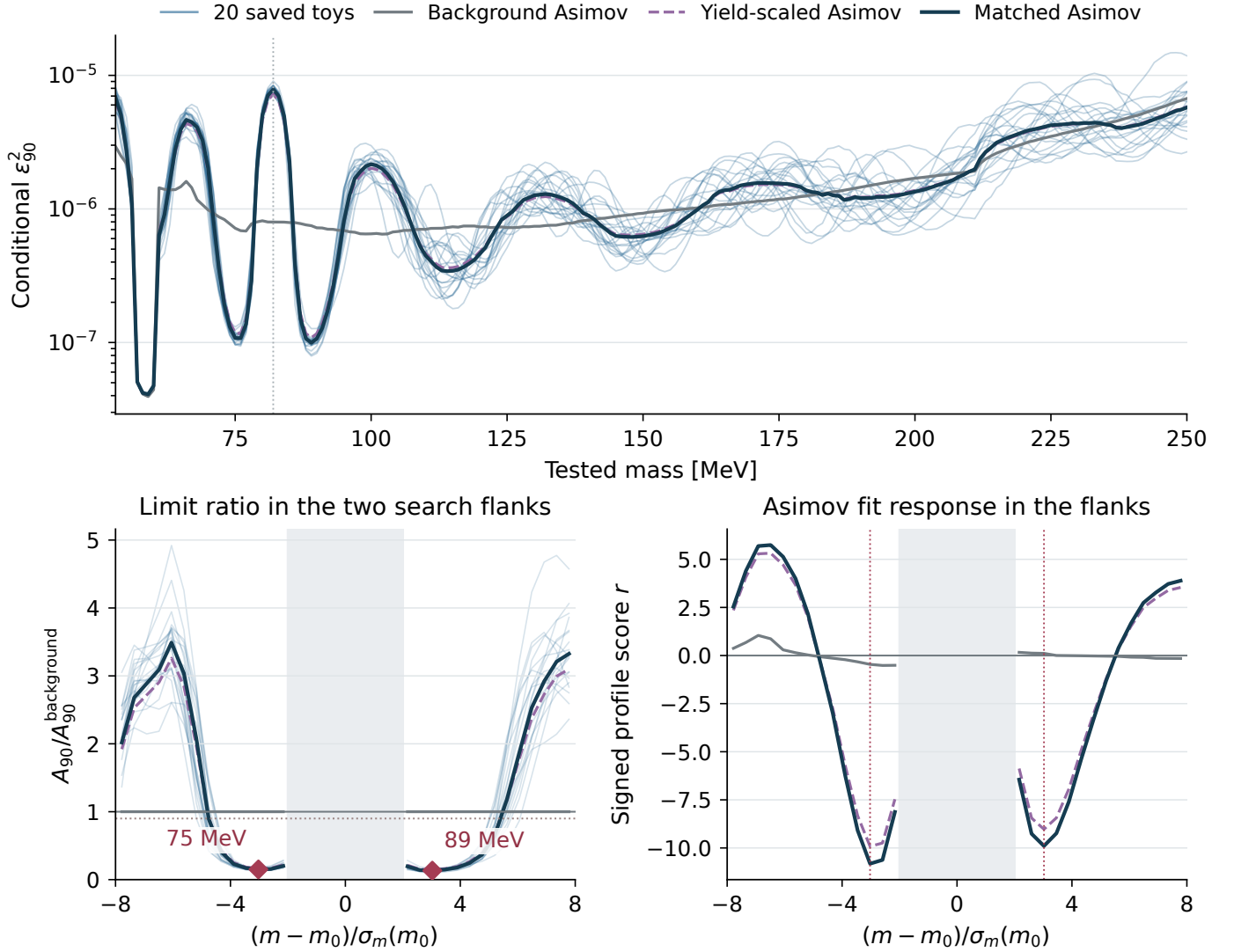
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	$60^\dagger$	79.2	58–60	60.0 [60.0, 60.0]	0.22 [0.20, 0.26]	20/20
Right	$70^\dagger$	74.0	70–73	70.0 [70.0, 70.0]	0.25 [0.23, 0.29]	20/20

Search bounds after clipping to the scan are left: 50.00–60.76 MeV; right: 69.24–81.97 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

75 MeV region: $1\% \times 100$

$m_0 = 82$ MeV, $\sigma_m = 2.313$ MeV; $Z_{\text{target}} = 15.32$, $a_{\text{match}}/a_{\text{yield}} = 1.095$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

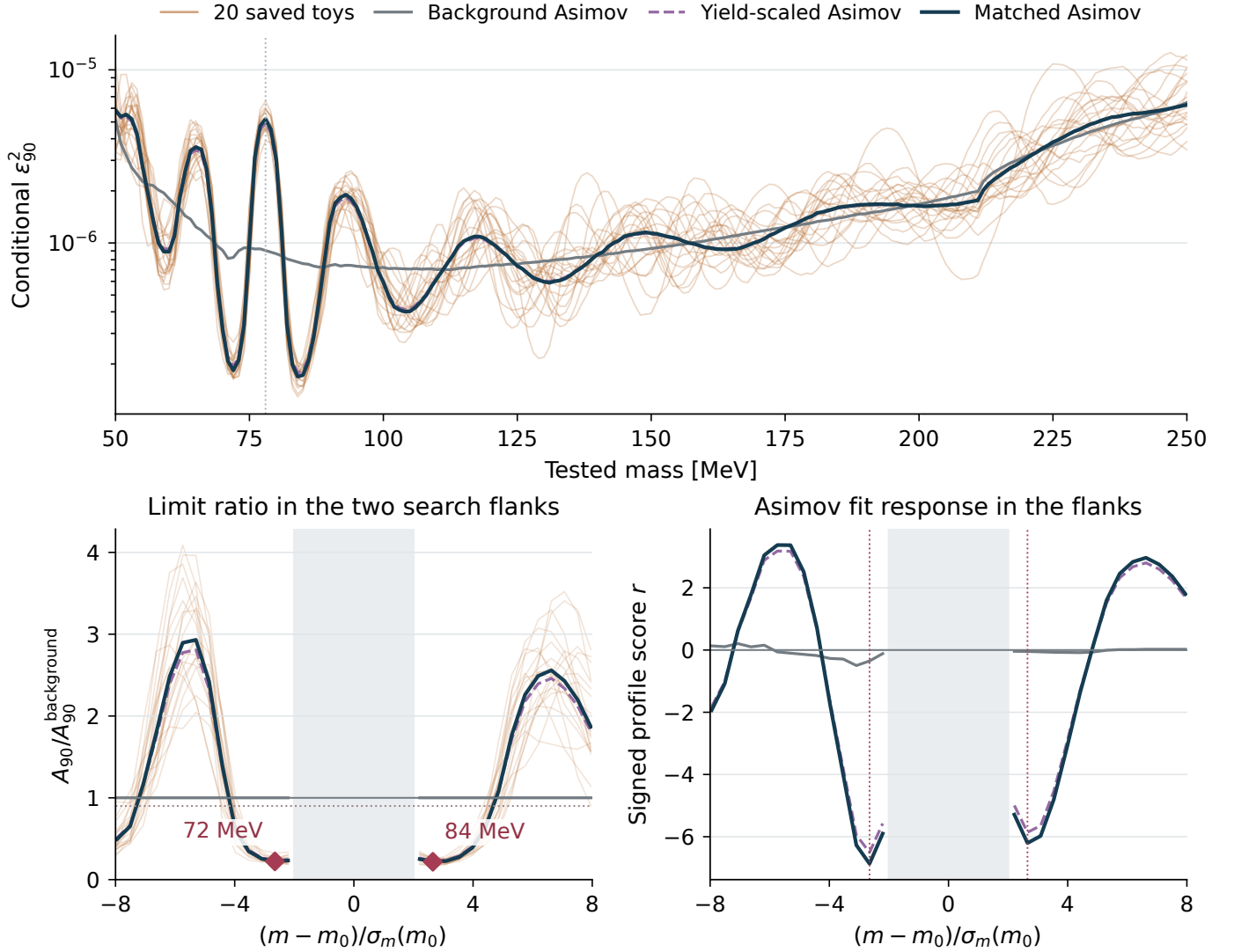
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	75	84.9	71–77	75.0 [75.0, 75.0]	0.15 [0.14, 0.16]	20/20
Right	89	86.7	87–94	89.0 [89.0, 89.0]	0.13 [0.13, 0.14]	20/20

Search bounds after clipping to the scan are left: 63.50–77.37 MeV; right: 86.63–100.50 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

75 MeV region: $10\% \times 10$

$m_0 = 78$ MeV, $\sigma_m = 2.263$ MeV; $Z_{\text{target}} = 8.88$, $a_{\text{match}}/a_{\text{yield}} = 1.059$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

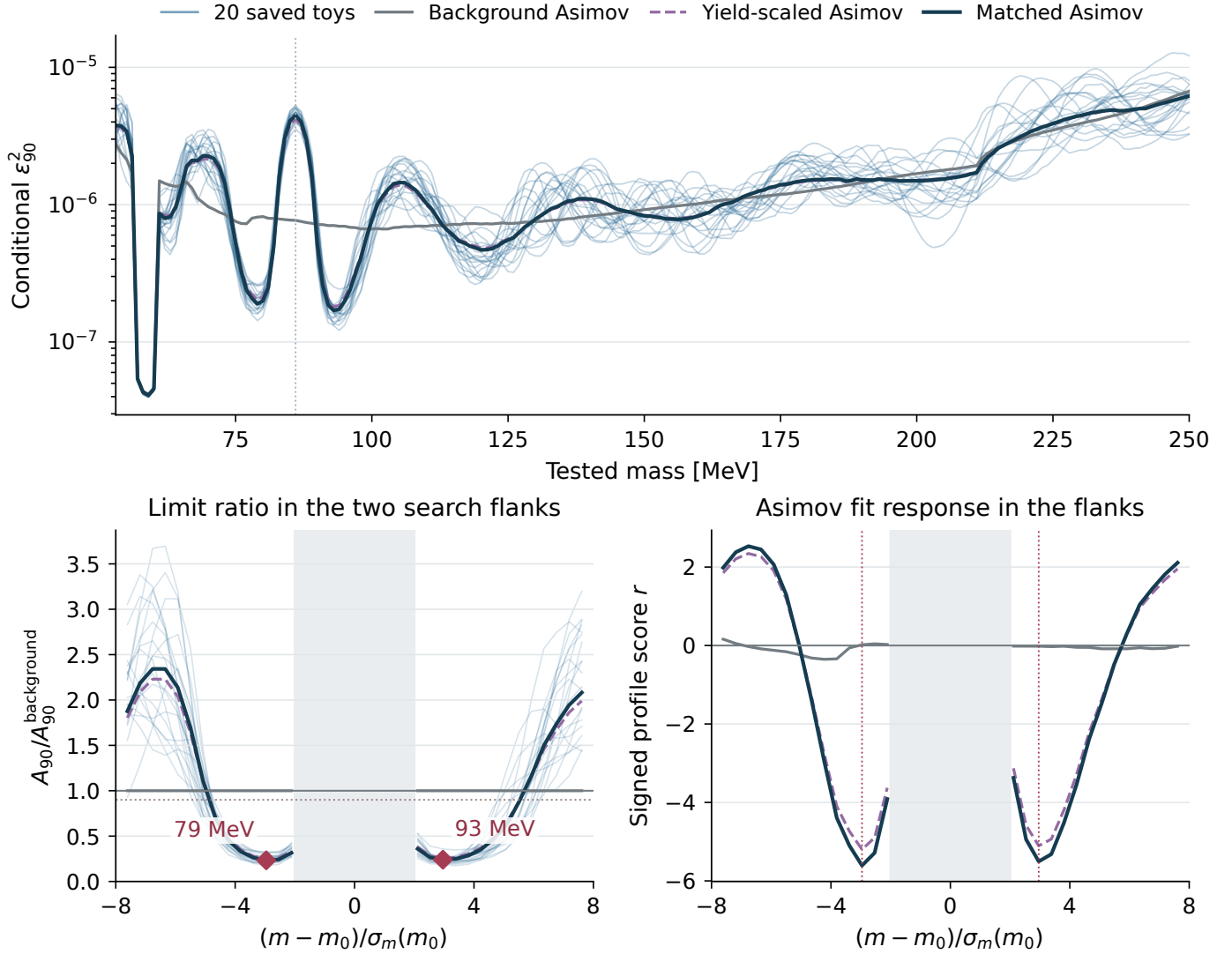
F flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	72	77.7	69–73	72.0 [72.0, 73.0]	0.22 [0.20, 0.26]	20/20
Right	84	78.2	83–88	84.0 [84.0, 85.0]	0.22 [0.19, 0.25]	20/20

Search bounds after clipping to the scan are left: 59.89–73.47 MeV; right: 82.53–96.11 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

92 MeV region: $1\% \times 100$

$m_0 = 86$ MeV, $\sigma_m = 2.365$ MeV; $Z_{\text{target}} = 8.61$, $a_{\text{match}}/a_{\text{yield}} = 1.078$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning. Inherited source-region boundary control.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

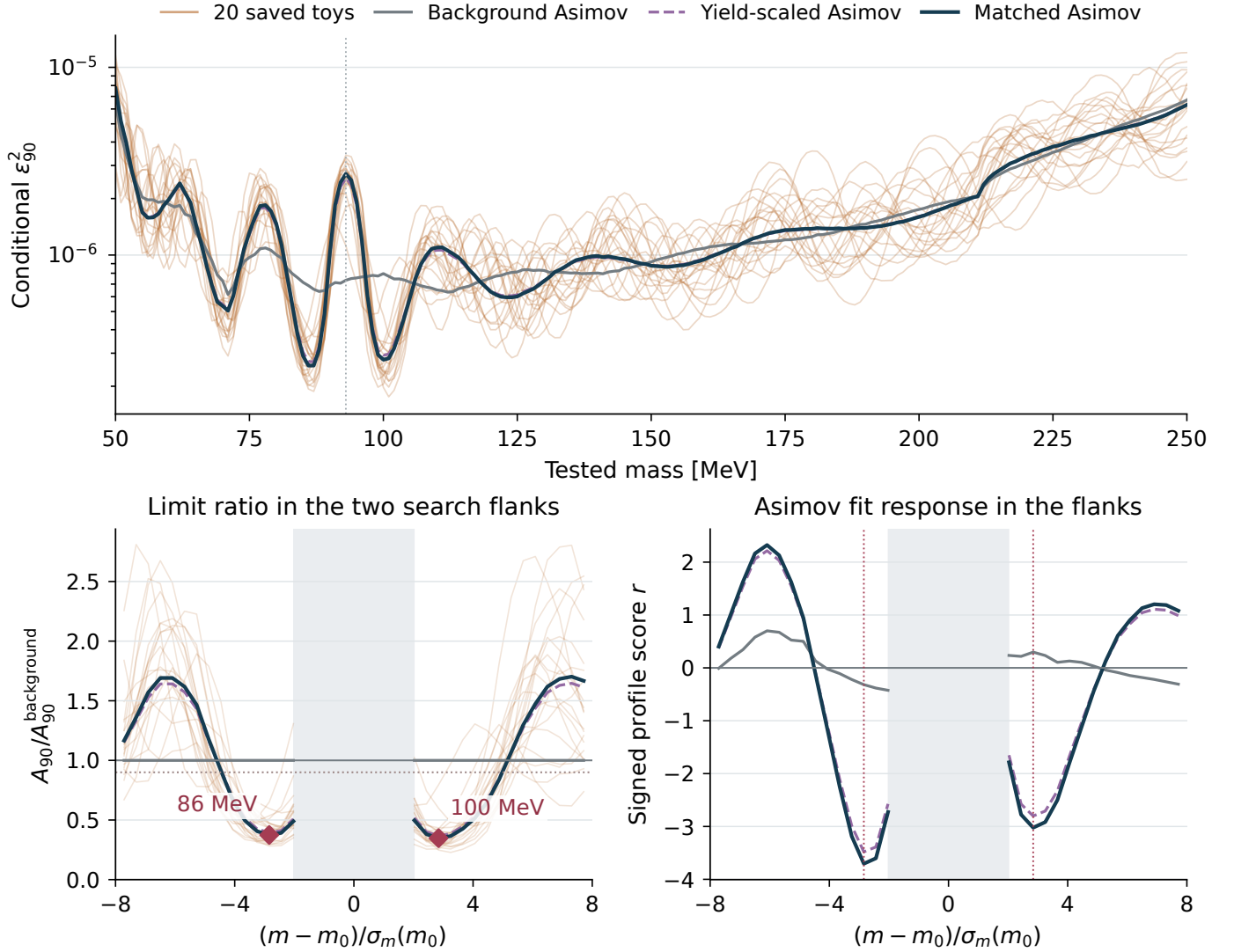
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	79	76.7	75–81	79.0 [79.0, 79.0]	0.24 [0.21, 0.29]	20/20
Right	93	76.0	91–99	93.5 [93.0, 94.0]	0.24 [0.20, 0.26]	20/20

Search bounds after clipping to the scan are left: 67.08–81.27 MeV; right: 90.73–104.92 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

92 MeV region: $10\% \times 10$

$m_0 = 93$ MeV, $\sigma_m = 2.463$ MeV; $Z_{\text{target}} = 4.77$, $a_{\text{match}}/a_{\text{yield}} = 1.071$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

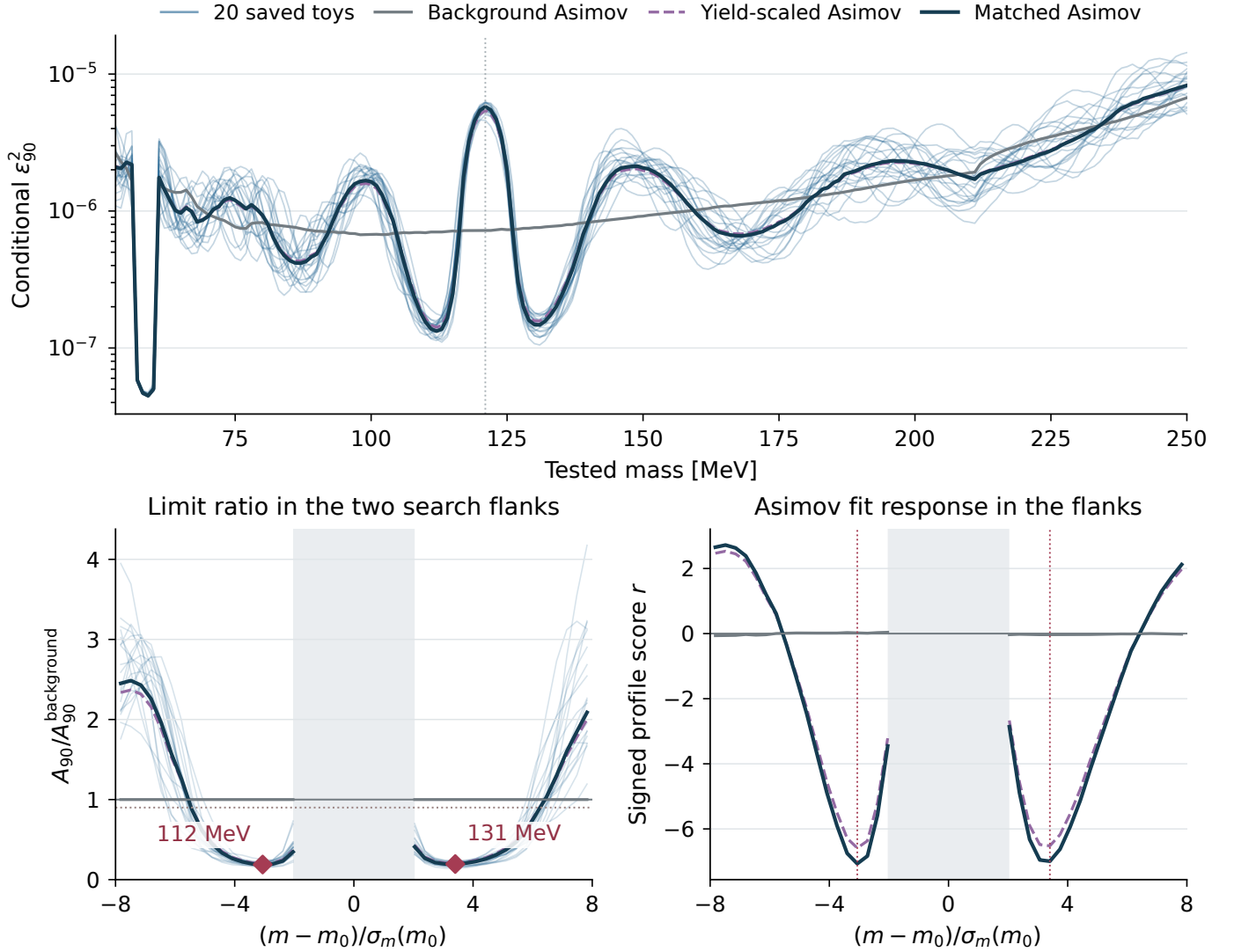
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	86	62.5	83–88	86.0 [85.0, 87.0]	0.41 [0.31, 0.51]	20/20
Right	100	65.2	98–105	100.0 [99.0, 101.0]	0.33 [0.29, 0.43]	20/20

Search bounds after clipping to the scan are left: 73.30–88.07 MeV; right: 97.93–112.70 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

120 MeV region: $1\% \times 100$

$m_0 = 121$ MeV, $\sigma_m = 2.939$ MeV; $Z_{\text{target}} = 11.94$, $a_{\text{match}}/a_{\text{yield}} = 1.073$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

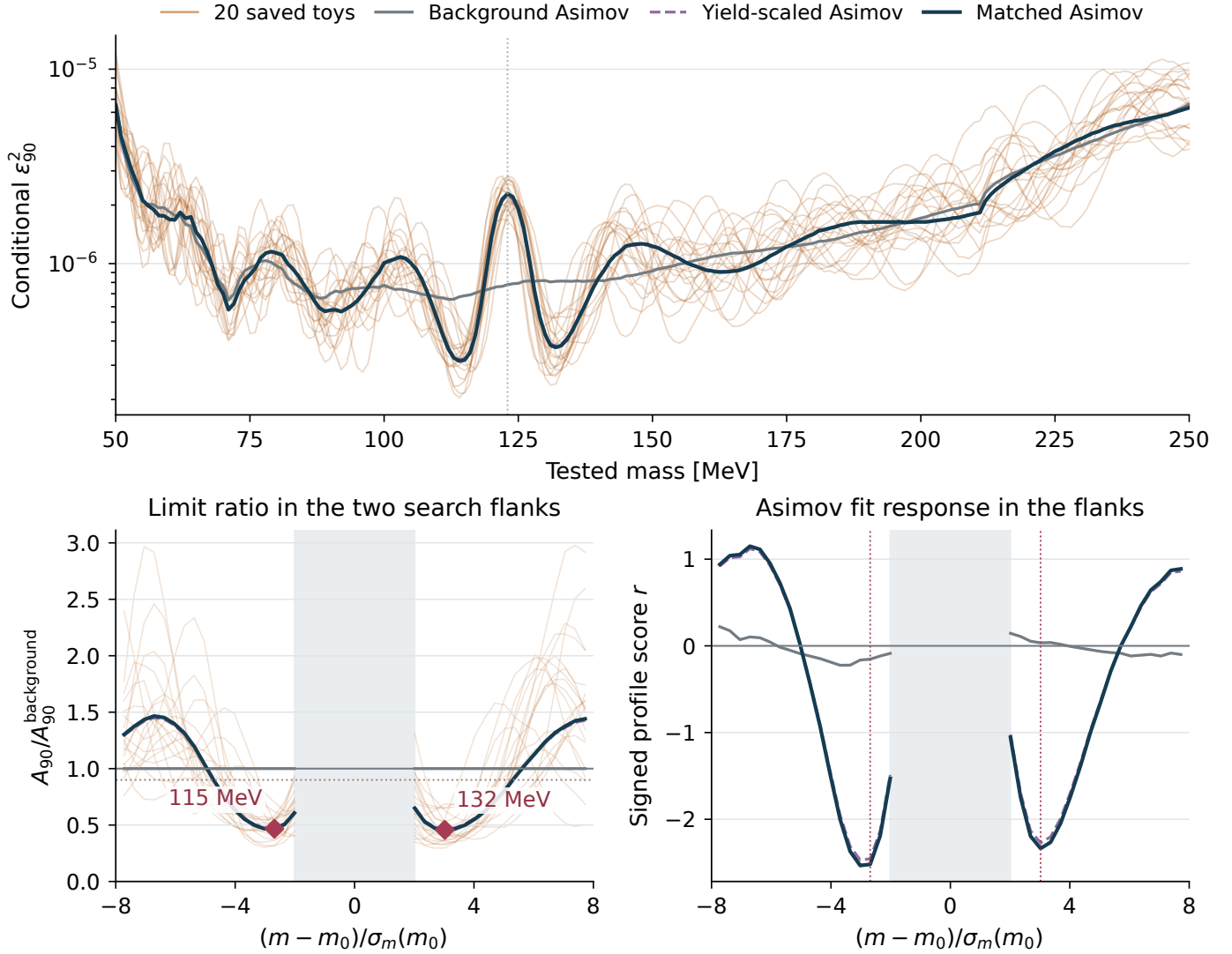
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	112	81.1	105–115	112.0 [111.0, 112.0]	0.20 [0.17, 0.22]	20/20
Right	131	80.5	127–139	131.0 [130.0, 132.0]	0.20 [0.16, 0.23]	20/20

Search bounds after clipping to the scan are left: 97.49–115.12 MeV; right: 126.88–144.51 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

120 MeV region: $10\% \times 10$

$m_0 = 123$ MeV, $\sigma_m = 2.978$ MeV; $Z_{\text{target}} = 3.74$, $a_{\text{match}}/a_{\text{yield}} = 1.028$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

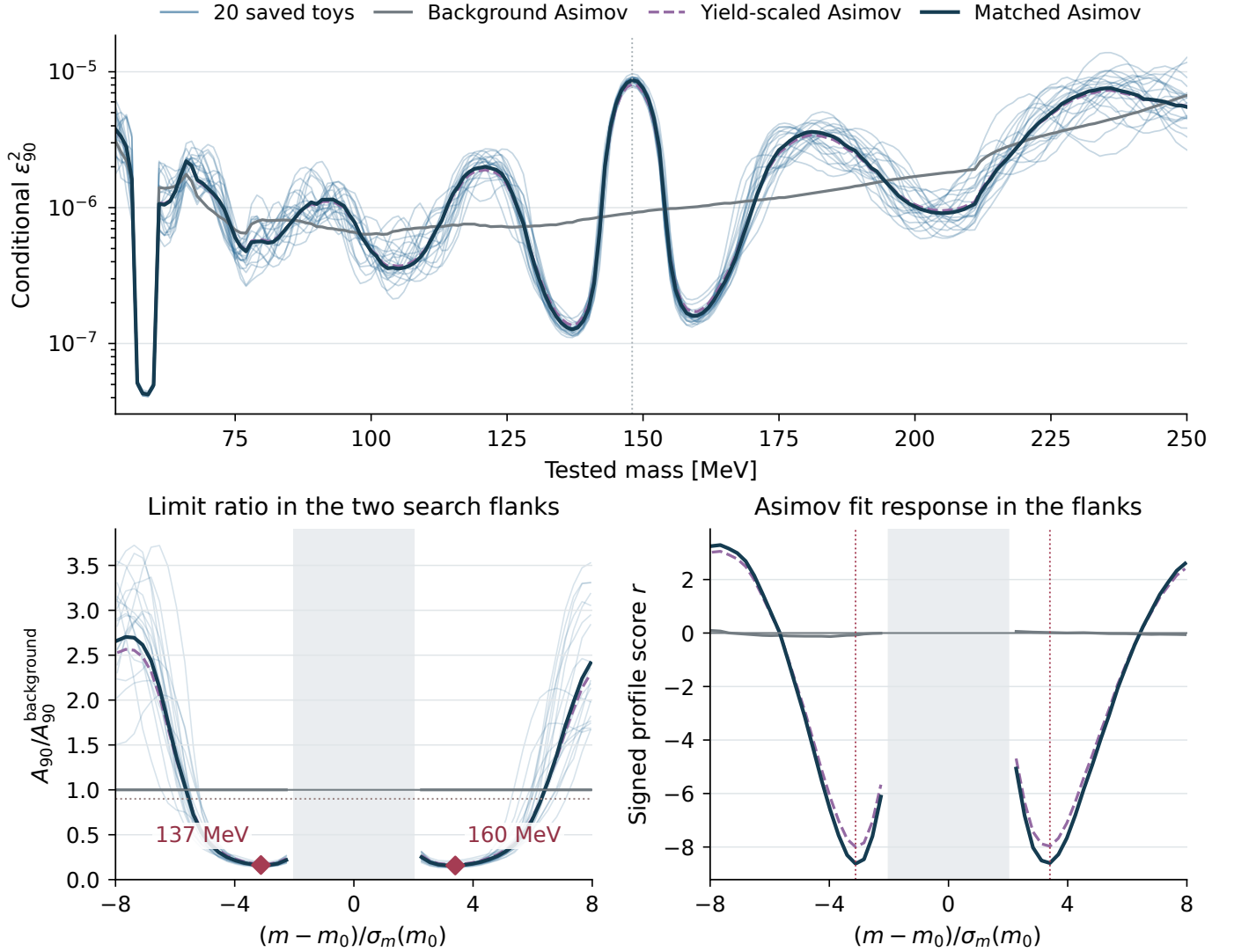
F flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	115	53.3	109–117	114.5 [112.1, 115.0]	0.51 [0.37, 0.56]	20/20
Right	132	54.3	129–138	132.0 [131.0, 133.0]	0.45 [0.35, 0.62]	20/20

Search bounds after clipping to the scan are left: 99.17–117.04 MeV; right: 128.96–146.83 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

160 MeV region: $1\% \times 100$

$m_0 = 148$ MeV, $\sigma_m = 3.526$ MeV; $Z_{\text{target}} = 14.66$, $a_{\text{match}}/a_{\text{yield}} = 1.079$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning. Inherited source-region boundary control.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

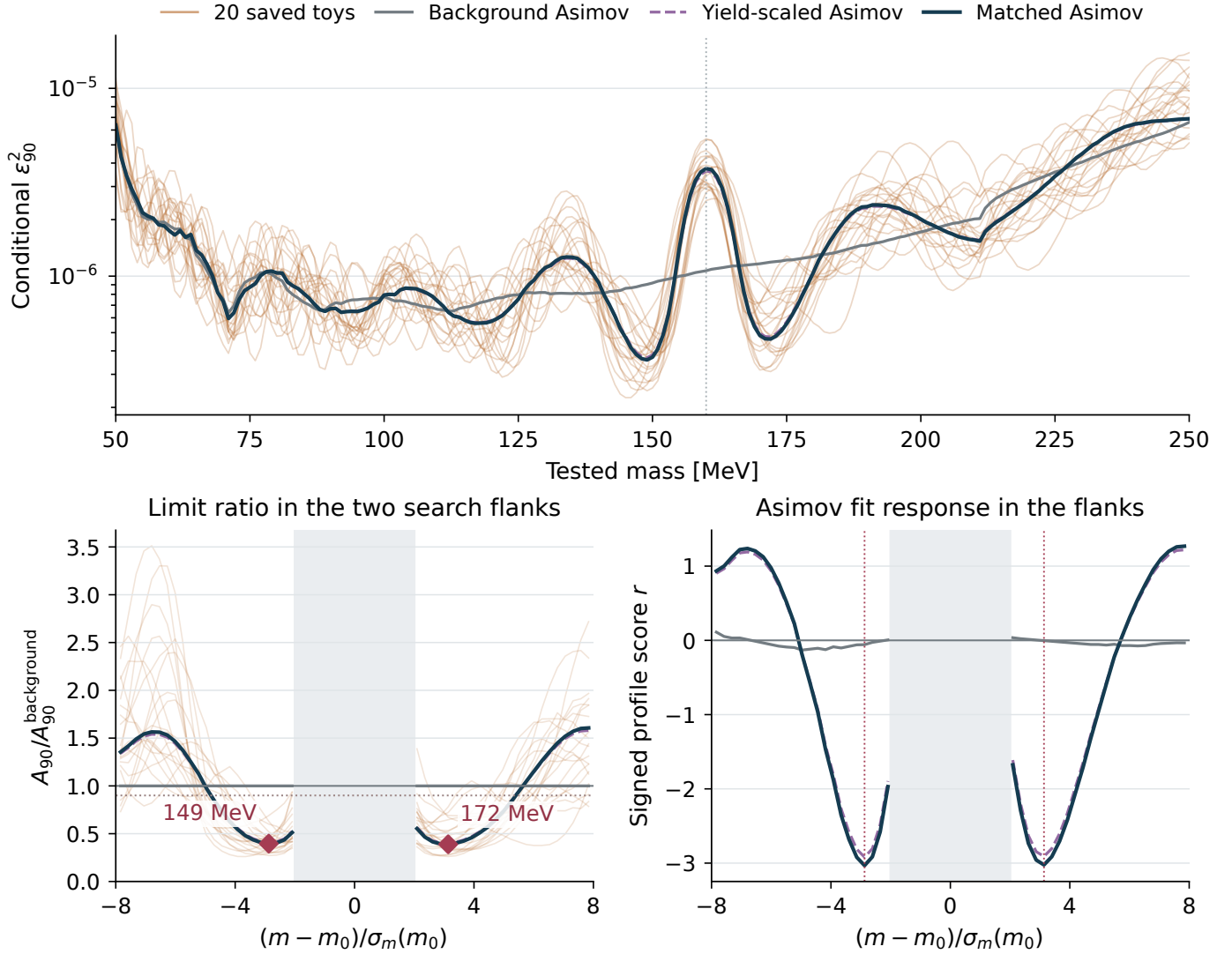
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	137	83.7	129–140	137.5 [137.0, 138.0]	0.16 [0.15, 0.18]	20/20
Right	160	84.2	156–170	159.5 [159.0, 160.0]	0.15 [0.14, 0.16]	20/20

Search bounds after clipping to the scan are left: 119.79–140.95 MeV; right: 155.05–176.21 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

160 MeV region: $10\% \times 10$

$m_0 = 160$ MeV, $\sigma_m = 3.827$ MeV; $Z_{\text{target}} = 4.71$, $a_{\text{match}}/a_{\text{yield}} = 1.039$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

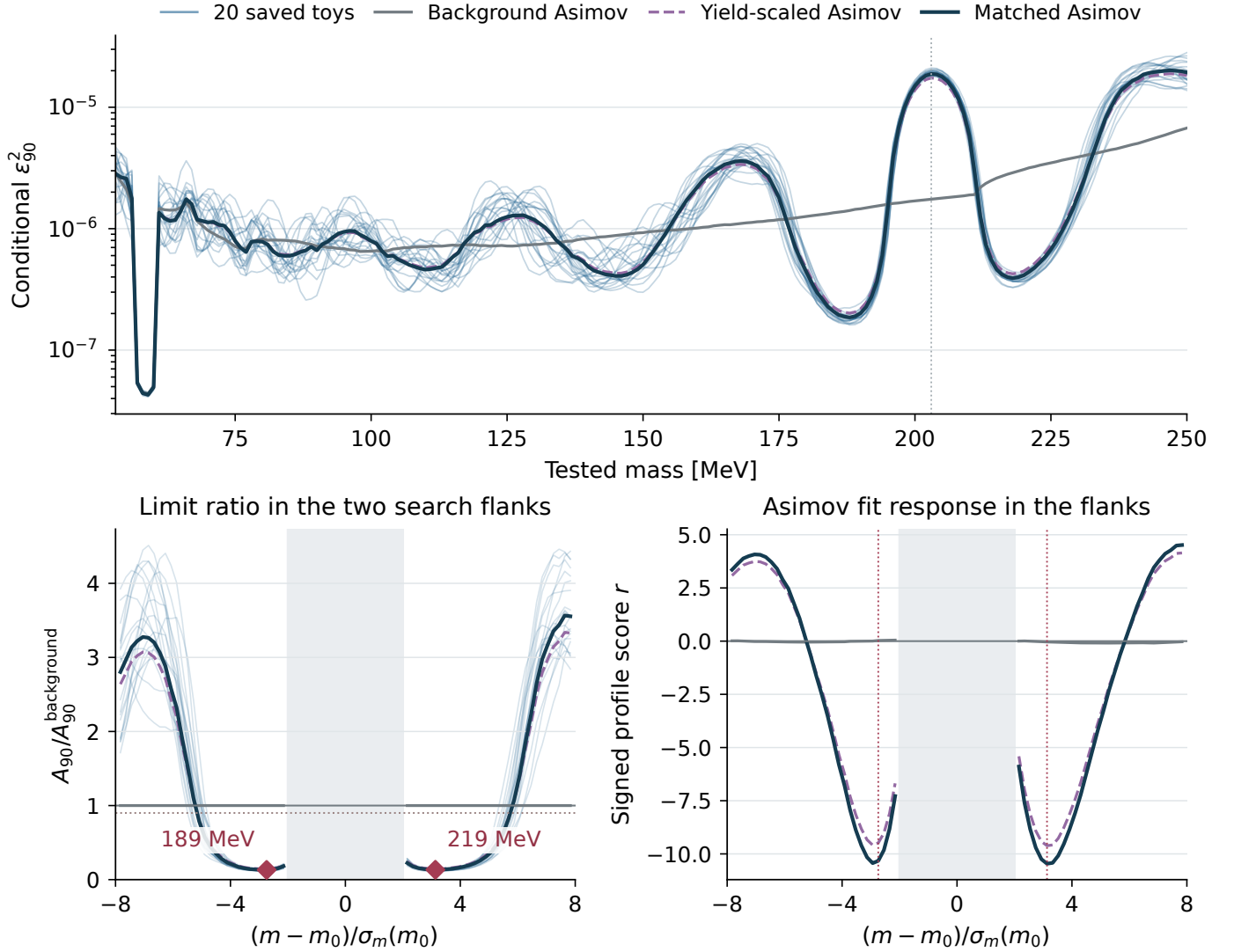
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	149	60.4	142–152	149.0 [146.0, 151.0]	0.43 [0.31, 0.52]	20/20
Right	172	61.1	168–180	172.0 [170.0, 173.0]	0.41 [0.33, 0.67]	20/20

Search bounds after clipping to the scan are left: 129.39–152.35 MeV; right: 167.65–190.61 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

210 MeV region: $1\% \times 100$

$m_0 = 203$ MeV, $\sigma_m = 5.108$ MeV; $Z_{\text{target}} = 16.87$, $a_{\text{match}}/a_{\text{yield}} = 1.089$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

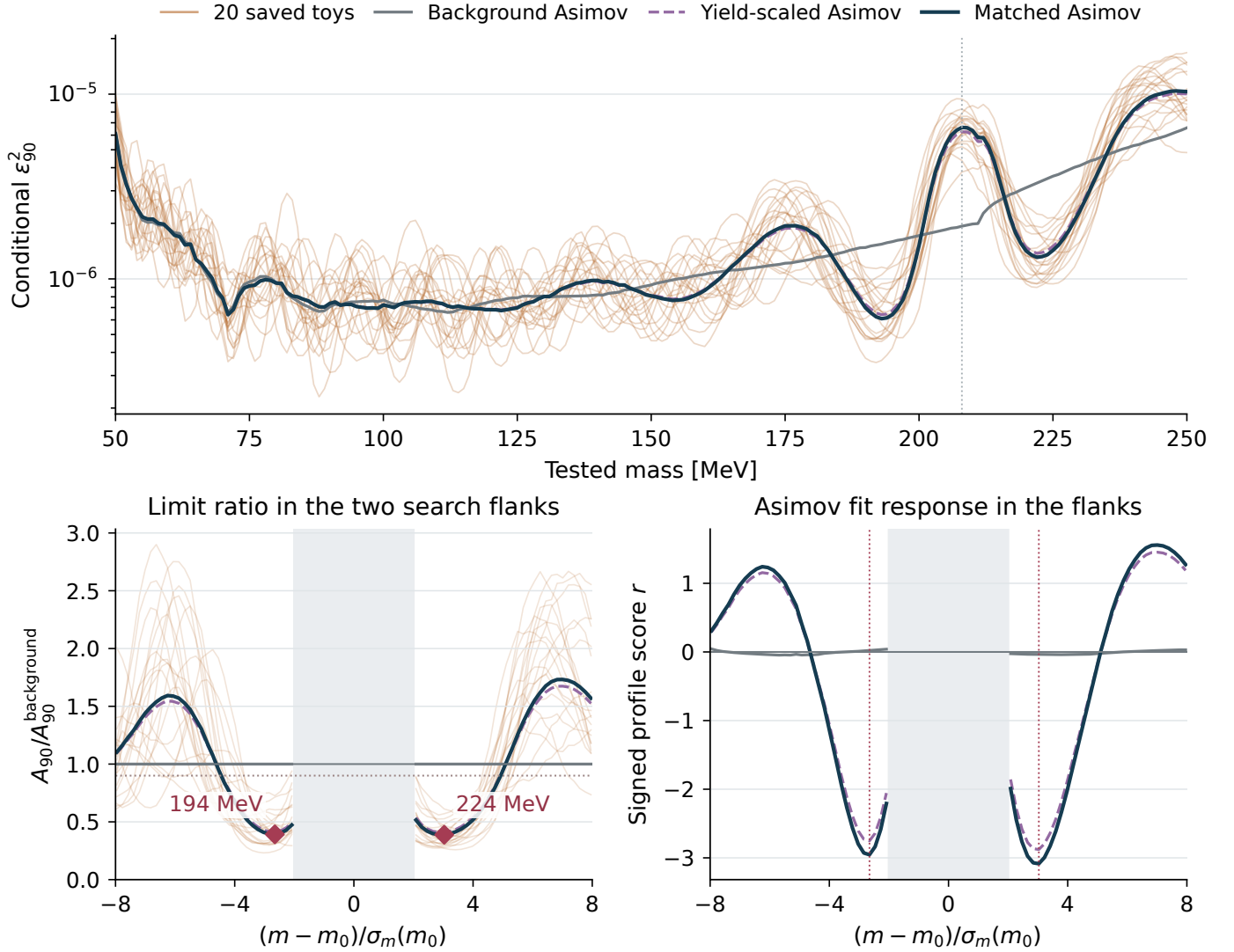
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	189	86.8	177–192	188.0 [188.0, 189.0]	0.13 [0.12, 0.14]	20/20
Right	219	86.6	214–232	219.0 [219.0, 220.0]	0.13 [0.12, 0.14]	20/20

Search bounds after clipping to the scan are left: 162.14–192.78 MeV; right: 213.22–243.86 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

210 MeV region: $10\% \times 10$

$m_0 = 208$ MeV, $\sigma_m = 5.278$ MeV; $Z_{\text{target}} = 4.46$, $a_{\text{match}}/a_{\text{yield}} = 1.073$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	194	60.8	185–197	193.0 [189.0, 195.0]	0.41 [0.32, 0.56]	20/20
Right	224	61.2	219–234	224.0 [222.0, 226.0]	0.37 [0.30, 0.48]	20/20

Search bounds after clipping to the scan are left: 165.78–197.44 MeV; right: 218.56–250.00 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

$m_0 = 84$ MeV, $\sigma_m = 2.339$ MeV; $Z_{\text{target}} = 15.33$, $a_{\text{match}}/a_{\text{yield}} = 1.092$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Figure 10 displays the Asimov fit response in the flanks, comparing four methods: 20 saved toys, Background Asimov, Yield-scaled Asimov, and Matched Asimov.

The top panel shows the conditional variance of the Asimov fit, ϵ_{90}^2 , as a function of the tested mass m [MeV]. The Matched Asimov method (dark blue line) shows the lowest variance, particularly in the signal region (shaded grey).

The bottom-left panel shows the limit ratio in the two search flanks, $A_{90}/A_{90}^{\text{background}}$, as a function of the normalized mass difference $(m - m_0)/\sigma_m(m_0)$. The Matched Asimov method (dark blue line) shows the lowest ratio, indicating the best fit response. Red diamonds mark the 78 MeV and 91 MeV points.

The bottom-right panel shows the signed profile score r in the flanks, as a function of the normalized mass difference $(m - m_0)/\sigma_m(m_0)$. The Matched Asimov method (dark blue line) shows a clear dip in the signed profile score in the signal region (shaded grey), indicating a strong fit response. Red dashed lines mark the 78 MeV and 91 MeV points.

Top: full-range coupling limits with each spectrum’s own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

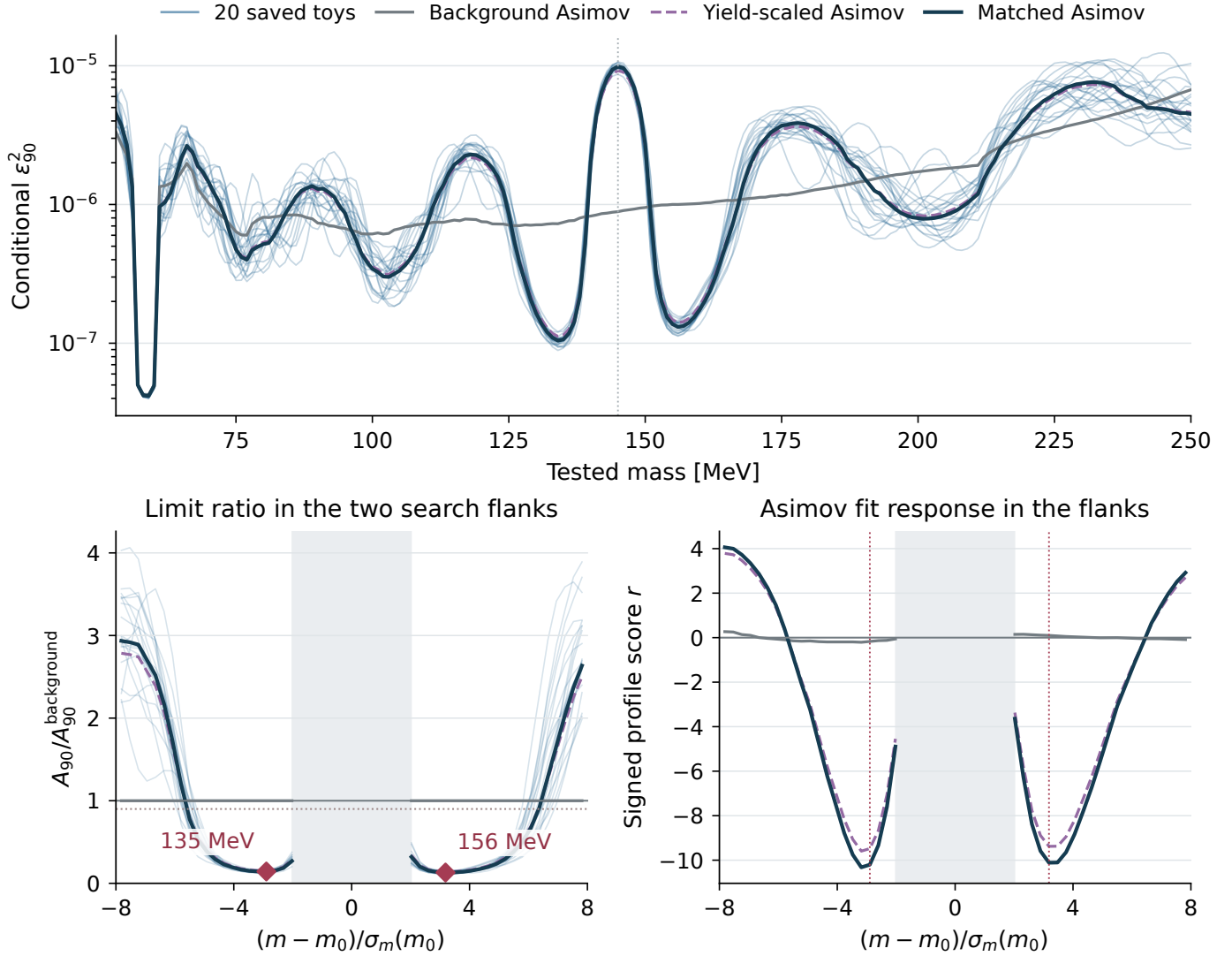
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	78	85.9	73–79	78.0 [78.0, 78.0]	0.14 [0.13, 0.15]	20/20
Right	91	86.1	89–96	91.0 [91.0, 91.0]	0.14 [0.13, 0.15]	20/20

Search bounds after clipping to the scan are left: 65.29–79.32 MeV; right: 88.68–102.71 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

Additional 1% injection: 145 MeV

$m_0 = 145$ MeV, $\sigma_m = 3.454$ MeV; $Z_{\text{target}} = 17.32$, $a_{\text{match}}/a_{\text{yield}} = 1.078$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

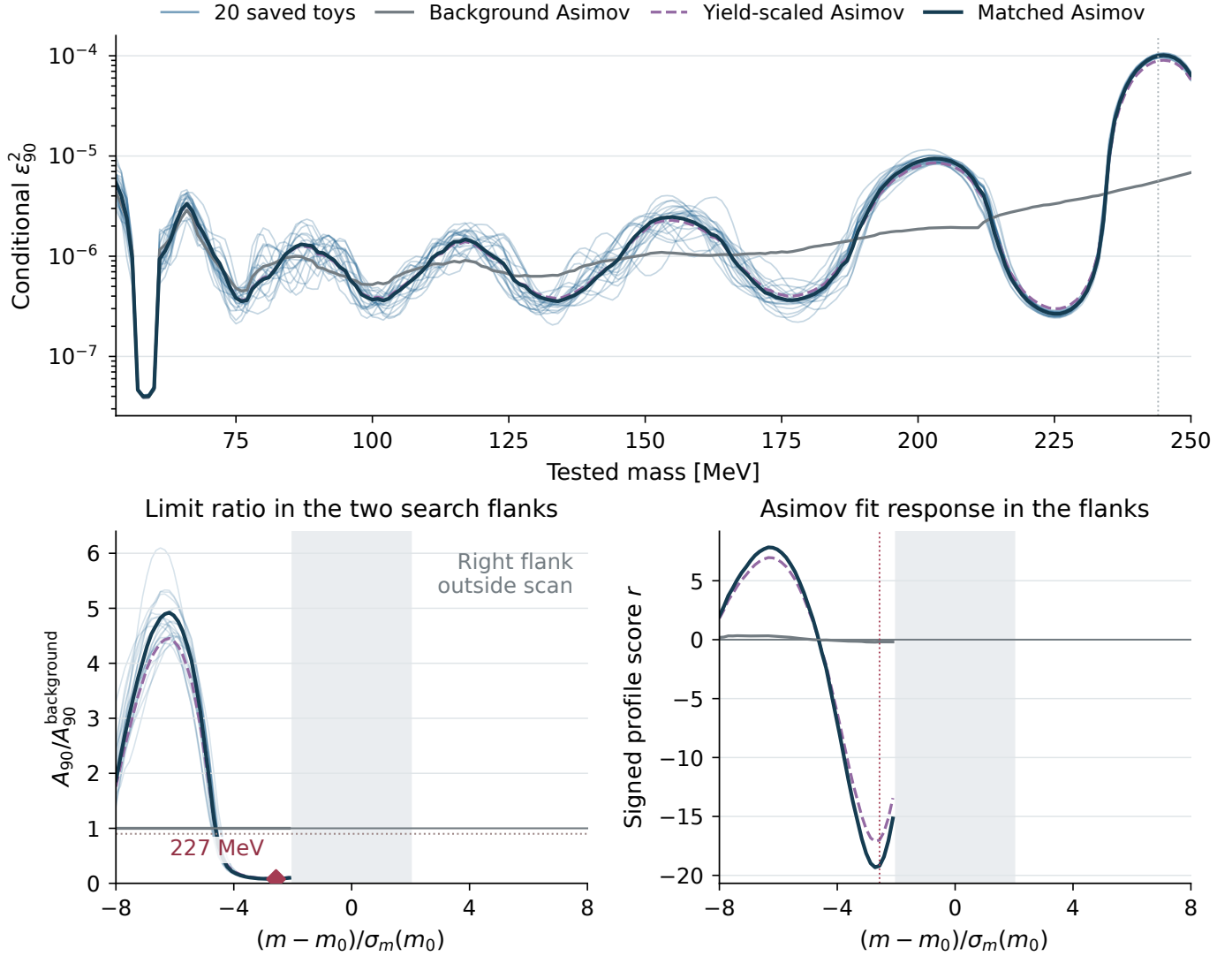
Flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	135	85.7	126–138	134.0 [134.0, 135.0]	0.14 [0.13, 0.15]	20/20
Right	156	86.9	152–166	156.0 [156.0, 157.0]	0.14 [0.12, 0.14]	20/20

Search bounds after clipping to the scan are left: 117.36–138.09 MeV; right: 151.91–172.64 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

Additional 1% injection: 244 MeV

$m_0 = 244$ MeV, $\sigma_m = 6.625$ MeV; $Z_{\text{target}} = 28.61$, $a_{\text{match}}/a_{\text{yield}} = 1.131$. The deterministic curves and all 20 saved toy limits use fresh GP conditioning.

Full scan: standard prompt-density conversion for each spectrum



Top: full-range coupling limits with each spectrum's own prompt density. Lower left: count-limit ratios; diamonds mark the selected matched-Asimov minima and the dotted line is $R = 0.90$. Lower right: signed Asimov profile scores with the same line styles. Gray central bands omit $|m - m_0| < 2\sigma_m$ from the two flank displays.

F flank	m_e [MeV]	Depth [%]	$R \leq 0.90$ [MeV]	Toy mass [MeV] median [16,84%]	Toy $R(m_e)$ median [16,84%]	$R \leq 0.9$ count
Left	227	92.2	214–230	226.0 [226.0, 227.0]	0.08 [0.07, 0.08]	20/20
Right	Search flank outside the saved mass scan.					

Search bounds after clipping to the scan are left: 191.00–230.75 MeV. † marks an endpoint and * a fallback minimum; the 1 MeV grid limits position precision.

Frozen forecast and reproducibility

This extension freezes the scan definitions and echo-selection rules before any comparison with observed 100% data. Its forecast concerns the response of the declared inference model to the saved v5.6 injection hypotheses. It is not an instruction to select a favorable observed minimum or to modify a subsequent scan.

Immutable parent inputs

The copied 300 toy histograms and all deterministic generator arrays retain their parent byte identities. The numerical engine, source-specific kernels and input catalogue are preserved alongside a copy manifest. The one-worker scan uses one linear-algebra thread and creates no new random samples.

Input	SHA-256
v5.6 source and injection catalogue	402697bffde75261a13d13eb9ea8c13f945c7450e2b9ce725ef2fc371d9b05d2
Extension protocol	1fff0feaf370dbdb83b19b8fb84bf978d65eaf9bf3e5048bfb77c21da76c2c61
Profiled limit solver	e6226d684a5e4129afdaf42fd5cc118f103b1793b39033f7b92d92a9ece0c3c5
Stable GP conditioning	f1bda485b2903fe778ff8c58841fb8cdac241ff3d63bb0ed6196fd0257ffe9e4

Saved numerical products

The extension contains 15 scenarios, 23 spectra per scenario and 68,724 mass-hypothesis rows. Each scan retains the fitted count limit, signed profile score, CL_s root, fit status, conditioning diagnostics, prompt density and both coupling-conversion variants.

Product	Contents
<code>inputs/toys/<scenario>.npz</code>	Saved v5.6 toy counts and generator arrays.
<code>derived/scans/<scenario>/*.csv</code>	Three deterministic and 20 toy full-grid scans.
<code>derived/scans/<scenario>/*.json</code>	Exact dependencies and corresponding CSV hashes.
<code>derived/echo_catalogue.csv</code>	Frozen selected flank minima and toy summaries.
<code>derived/toy_echo_locations.csv</code>	Each toy's selected mass and ratio at the fixed Asimov echo.
<code>derived/report_display_ledger.json</code>	Display completeness and conversion controls.
<code>inputs/copy_manifest.json</code>	Provenance and byte identities of copied inputs.
<code>qa/</code>	Numerical, provenance and rendered-page validation records.

Numerical safeguards and rebuilding

Validation passed all 2,847 numerical checks and all 756 echo-table integrity checks. The log-GP is conditioned in the positive eigenspace of the RBF covariance at a relative cutoff of 10^{-14} , using an SVD ridge calculation to avoid subtraction of nearly equal posterior covariances. The profiled solver requires positive fitted means and a converged bounded CL_s root. Per-spectrum sideband prediction, covariance and Poisson noise are recomputed at fixed archived kernel coordinates. Kernel optimization variability and hyperparameter uncertainty are not sampled. From the package root, `pythonscripts/scan_limits.pyrun` resumes the saved scans after checking their dependencies. `pythonscripts/summarize_echoes.py` builds the selected-minimum ledger. Then `pythonscripts/make_report.py` regenerates figures and `source/main.tex`; compile that file from its own directory. The report generator performs no fits or random draws. Numerical checks and deterministic reproducibility establish that the stated model was evaluated consistently. They do not establish calibrated coverage, rare-tail probabilities or fidelity to unobserved full-data structure.

References

- [1] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C **71**, 1554 (2011); erratum **73**, 2501 (2013). arXiv:1007.1727. Bounded upper-limit statistics and Asimov distributions provide the asymptotic conventions.